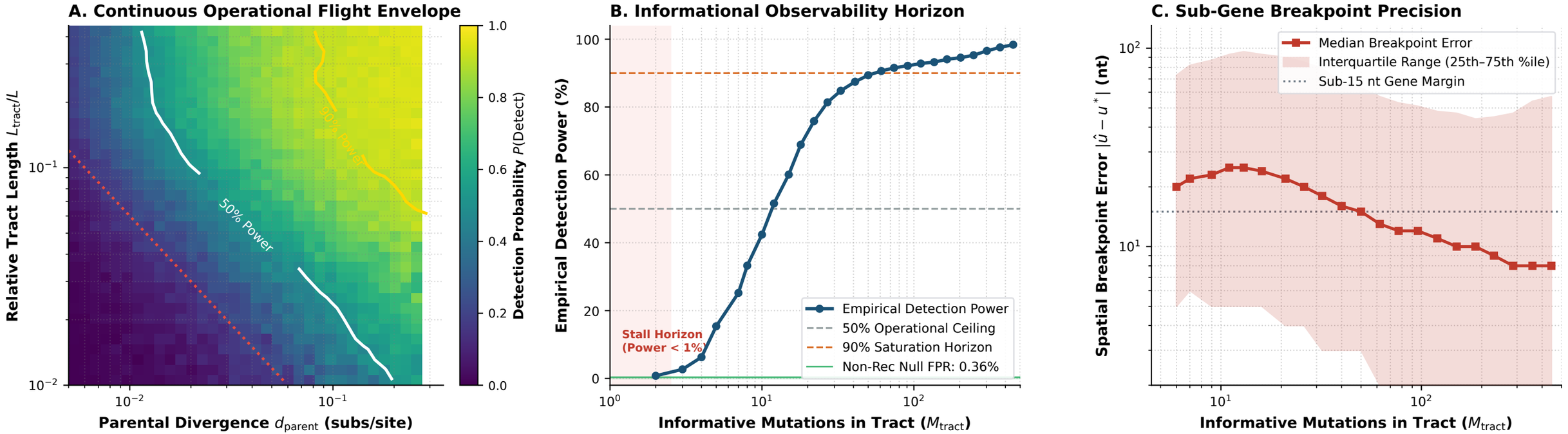

A. Continuous Operational Flight Envelope
Relative Tract Length L_{tract}/L
10^{-1}
10^{-2}
90% Power
50% Power
10^{-2}
10^{-1}
Parental Divergence d_{parent} (subs/site)
1.0
0.8
0.6
0.4
0.2
0.0
Detection Probability P(Detect)
B. Informational Observability Horizon
Empirical Detection Power (%)
100
80
60
40
20
0
Stall Horizon
(Power < 1%)
Empirical Detection Power
50% Operational Ceiling
90% Saturation Horizon
Non-Rec Null FPR: 0.36%
10^0
10^1
10^2
Informative Mutations in Tract (M_{tract})
C. Sub-Gene Breakpoint Precision
Spatial Breakpoint Error $|\hat{u} - u^*|$ (nt)
10^2
10^1
Median Breakpoint Error
Interquartile Range (25th–75th %ile)
Sub-15 nt Gene Margin
10^1
10^2
Informative Mutations in Tract (M_{tract})